Finding landslide scars from satellite data in Pacitan

A screening model for Cyclone Cempaka (Nov 2017): what it finds, what it misses, and what it hands to an expert

Model report, November 2026

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The model finds 86.8% of scars but only 9.0% of its flags are real

Answer. On 106 scars of 500 m2 or more in held-out blocks, the model detects 86.8% (target 80%: met). Of the objects it flags, 9.0% touch a scar (target 30%: not met). The criterion stays as written.

Why it still helps. An expert checks 10 false objects per true scar instead of 31 with the baselines, over 1.0% of the area.

What the false alarms are. In a blinded review of 30 large false alarms, 1 was judged a scar. Most look like terraces, river banks or roads, not unmapped landslides.

Use. A screening aid, not a map of every landslide. 5,865 candidates are delivered for expert verdicts, the 1,678 large ones first.

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The goal: flag candidate scars for expert review, with recall of at least 80% and precision of at least 30%

Recall counts inventory polygons found; precision counts flagged objects that touch a polygon.

Aim. A screening aid that flags candidate landslide scars from satellite data, so an expert reviews flags instead of digitizing the whole area by hand.

Event and place. Cyclone Cempaka, November 2017, Pacitan regency, East Java: a 1,370 km2 study domain.

Success, fixed before any model. Object recall of 80% or more on polygons of 500 m2 or more (5 pixels), at object precision of 30% or more, on held-out 3 km blocks.

Why precision is a floor. The inventory is not wall-to-wall: a flag with no polygon under it may still be a real scar that nobody mapped.

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Recall: how many real scars did it find?

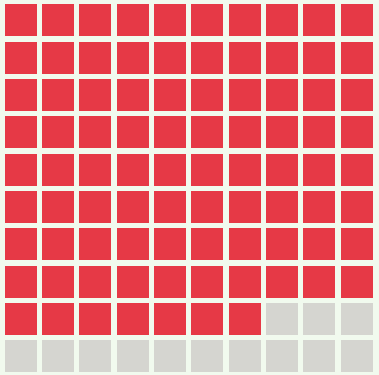
RECALL

Question. Out of every 100 real scars of 500 m2 or more, how many did the model find?

Counted as found. At least one flagged object touches the scar.

On the test blocks. 92 of 106 scars found: 86.8%. Target 80% or more: met.

Read it with care. Flagging more of the map raises recall, so it is never read alone.



87

scars found, out of every 100 real scars

Precision: how many flags are real?

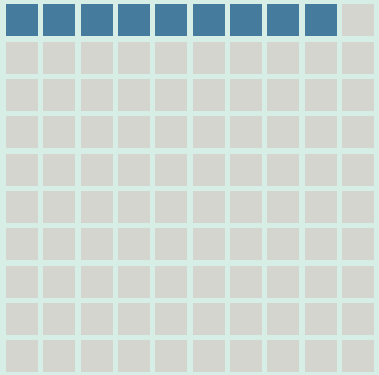
PRECISION

Question. Out of every 100 flagged objects, how many touch a real scar?

Counted as real. The object touches a scar mapped in the inventory.

On the test blocks. 105 of 1,170 objects touch a scar: 9.0%. Target 30% or more: not met.

A floor. A flag on a real scar that nobody mapped counts as wrong, so true precision is higher.



9

flags that are real, out of every 100 flagged objects

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Labels are 743 polygons from a citizen-report inventory, so a missing polygon is not proof of no landslide

Observed: labels and imagery. Assumed: the shift, and that unmapped ground is mostly no landslide.

Labels. Samodra et al. 2024 inventory (CC BY 4.0), built from 328 citizen reports extended with 50 cm imagery and field checks. It states no footprint.

Inputs. Sentinel-2 satellite images before and after the storm, a 2018 satellite embedding, and the release's terrain, land-use and geology maps, all on one 10 m grid. The next page lists the 97 features.

Rare and small. Scars cover 0.15% of the area; the median polygon is 1,096 m2 and 26% are under 500 m2, below what 10 m pixels show reliably.

Registration. Polygons sit about 12.6 m off the Sentinel-2 grid (found from the data, cause unverified) and were shifted onto it; predictions shift back to compare with the inventory.

Samodra et al. 2024, Mendeley Data 10.17632/vrtx3w6mjd.1; Sentinel-2: Copernicus; embedding: Google DeepMind (CC BY 4.0)
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Each 10 m pixel is described by 97 numbers in six groups, from vegetation change to terrain and a satellite embedding

NDVI is a vegetation index from satellite light: high is dense green, low is bare ground. dB is its change after the storm.

Vegetation at the pixel (5). Greenness before the storm (pre), after it (postA, postB) and the change between them (dA, dB).

Vegetation around the pixel (14). Mean, lowest value and spread of the change within 30 m and 50 m (dB_min5 is the lowest within 50 m), and how the pixel differs from its 110 m surroundings.

Terrain (10). Slope, elevation, distance to river and road, wetness index (twi), stream power (spi), curvature and aspect.

Land use and lithology (2). Class codes from the release's land-use and geology maps. With terrain they make up the 'site maps' of the model comparison.

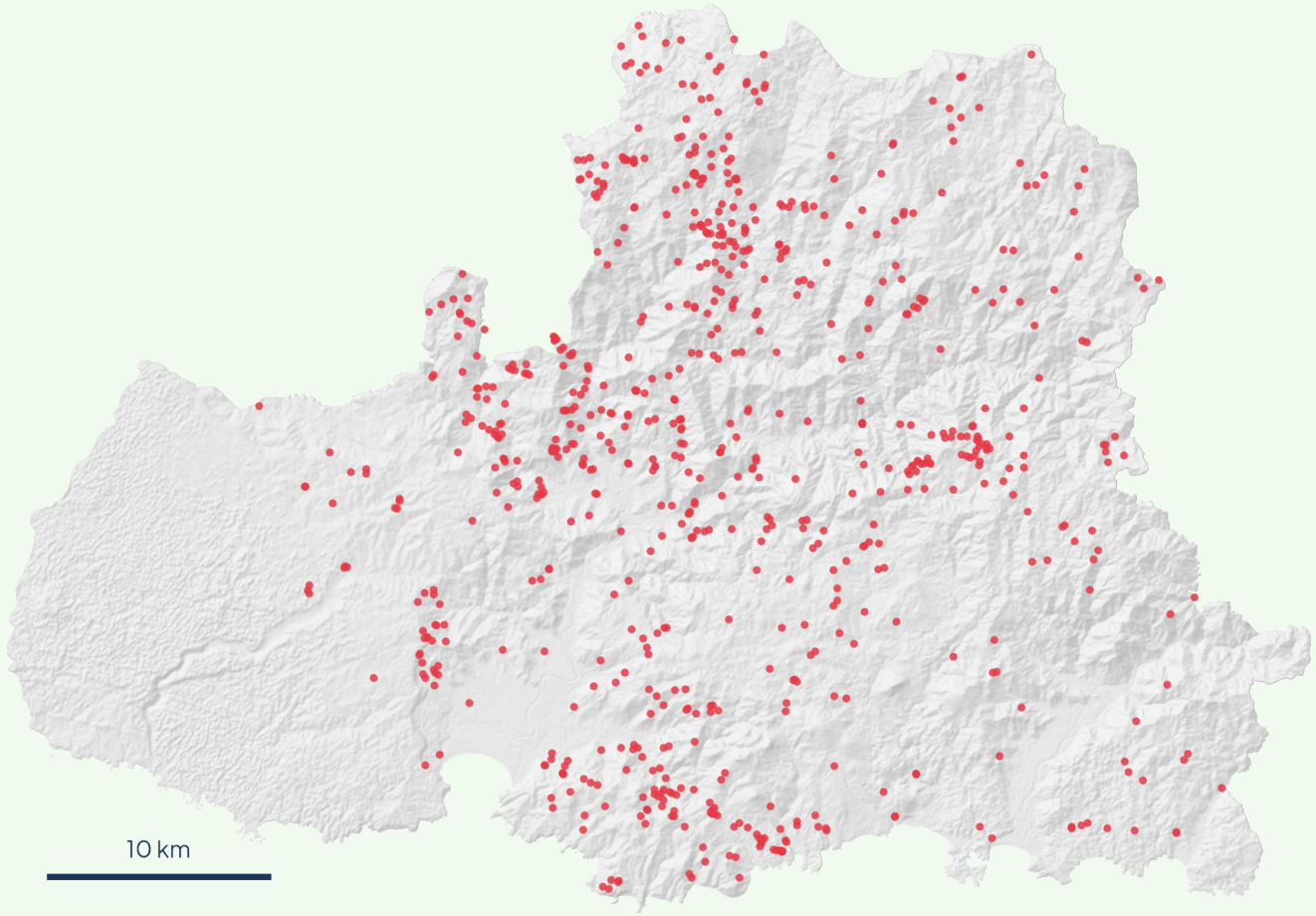
Satellite Embedding (64). A 64-number summary of what the satellite saw in 2018 at that pixel, from a Google model (e18_A00 to e18_A63).

Missing-data flags (2). Mark pixels with no cloud-free observation or no wetness index.

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Scars sit on steeper ground: the median slope at a scar is 22 degrees, against 15 across the study area

The map shows where scars were mapped, not where they happened: the inventory started from citizen reports.



- inventory polygon, centroid (n = 743)

What is drawn. Centroids of the 743 inventory polygons on a hillshade of the release's 10 m elevation model. UTM 49S, north up.

Steeper ground. 37% of scars sit on slopes of 25 degrees or more; 16% of the study area does.

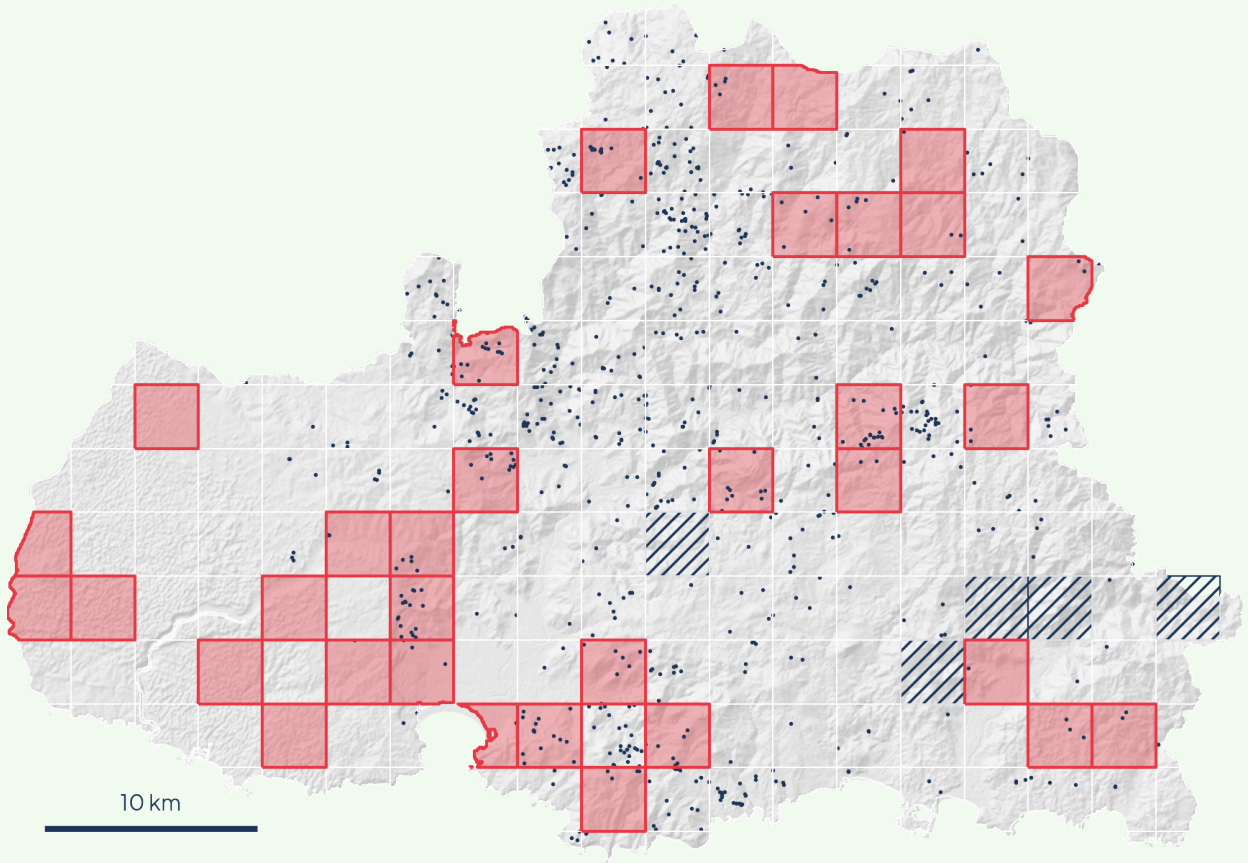
Gaps. Few scars are mapped on the flat western plain. A gap is not proof of no landslide.

Samodra et al. 2024 (CC BY 4.0): inventory and elevation model; slope read from the release's slope layer

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Scores come from whole 3 km blocks held out of training, so spatial neighbours cannot leak into the test

34 of 170 blocks (seed 42) hold 157 of 743 polygons and are scored once. The score on page 11 is one run of a frozen model and threshold.



- held-out test block (34)
- training block (white edges)
- excluded as likely unmapped (5)
- inventory polygon, centroid

Model choice. Repeated 5-fold block cross-validation, 5 seeds; models compared at 80% object recall.

Why not AUC. AUC, a ranking score where 0.5 is a coin flip, hides the false alarms of a rare event; random folds also flatter average precision by up to 0.06.

Excluded. Five blocks look unmapped and stay out of the domain.

Intervals. Block bootstrap 95%; 106 test polygons of 500 m2 or more, so intervals are wide.

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A plain vegetation-loss threshold needs 1.08% of the area to reach 50% recall, and only 7% of those pixels are scars

A threshold alone is a weak detector, so the other features have to remove false alarms.

Pixel recall	Threshold on NDVI loss	Pixel precision	Flagged pixels	Area flagged
20%	0.292	19.4%	16,675	0.16%
50%	**0.111**	**7.0%**	**115,502**	**1.08%**
80%	-0.051	0.6%	2,034,645	19.01%

Note: NDVI loss = pre-event NDVI minus April-May 2018 NDVI; pixel-level; recall of 80% flags a fifth of the area

Source: reports/tables/eda_naive_detector.csv

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Vegetation change and the satellite embedding complement each other: together 7.6% precision at 80% recall, against 5.0% and 3.7% alone

Cross-validated on training blocks, 5 repeats. The intervals overlap; the paired comparison is in the report.

Model	Precision at 80% recall	95% interval	Area flagged
All features (boosting), chosen	**7.6%**	**3.9% to 13.7%**	**1.0%**
Vegetation change + site maps (boosting)	5.0%	2.7% to 9.4%	1.8%
Satellite embedding + site maps (boosting)	3.7%	2.8% to 5.3%	1.2%
All features (logistic)	4.1%	2.8% to 7.8%	2.0%
Simple logistic (vegetation change + site maps)	2.8%	1.9% to 5.7%	4.0%
Naive threshold on vegetation loss	2.7%	2.2% to 3.8%	12%

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Recall target met at 86.8%, precision target not met at 9.0%; the criterion stays unchanged

Frozen model scored once on 34 held-out blocks: 1,170 objects flagged, 105 touching a scar.

86.8%

object recall, 95% interval 79.0% to 93.1% (target 80%, met)

9.0%

object precision, 95% interval 6.7% to 11.5% (target 30%, not met)

10

false objects to check per true scar (baselines: 31)

1.0%

of the test area flagged for review

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At similar recall the model flags about a third as many objects as the baselines and recovers scar shapes better

Same 34 test blocks, 157 polygons (106 of 500 m2 or more). Intervals overlap on recall; they do not on precision.

Model	Object recall	Strict recall	Object precision	Objects flagged
Chosen model (boosting)	**86.8% (79.0–93.1)**	**69.8%**	**9.0% (6.7–11.5)**	**1,170**
Simple logistic (vegetation change + site maps)	89.6% (82.9–95.3)	50.9%	3.1% (2.2–4.1)	3,569
Naive threshold on vegetation loss	89.6% (81.8–95.9)	34.0%	3.2% (2.1–4.4)	3,821

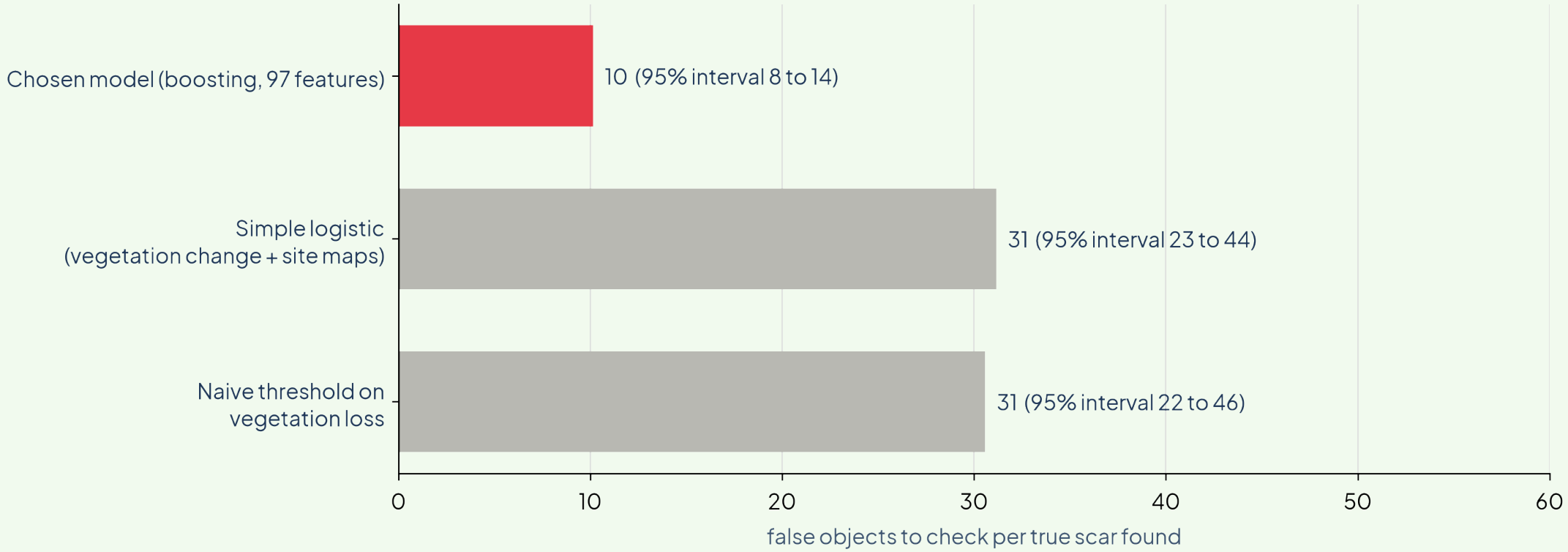
Note: strict recall needs overlap with a polygon (IoU 0.3); the loose 'touches' rule favours long objects

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About 10 false objects to check per true scar, against 31 for the baselines

The review workload is what an expert pays for each scar found.

False objects per true scar found, test blocks, 95% interval



Source: reports/tables/test_results.csv

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Reaching 30% precision would cost recall: on the test blocks it appears only at about 68%

The curve is descriptive (106 polygons); the operating point was frozen from training-block repeats, not read off this curve.

Object precision against object recall, test blocks, model sweep over thresholds



Source: reports/tables/test_curve.csv, test_results.csv

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Misses are the small scars: 98% of polygons over 20 pixels are found, against 35% of those under 5

Share of polygons touched by a flagged object. The baselines find more small ones, at about three times the review workload.

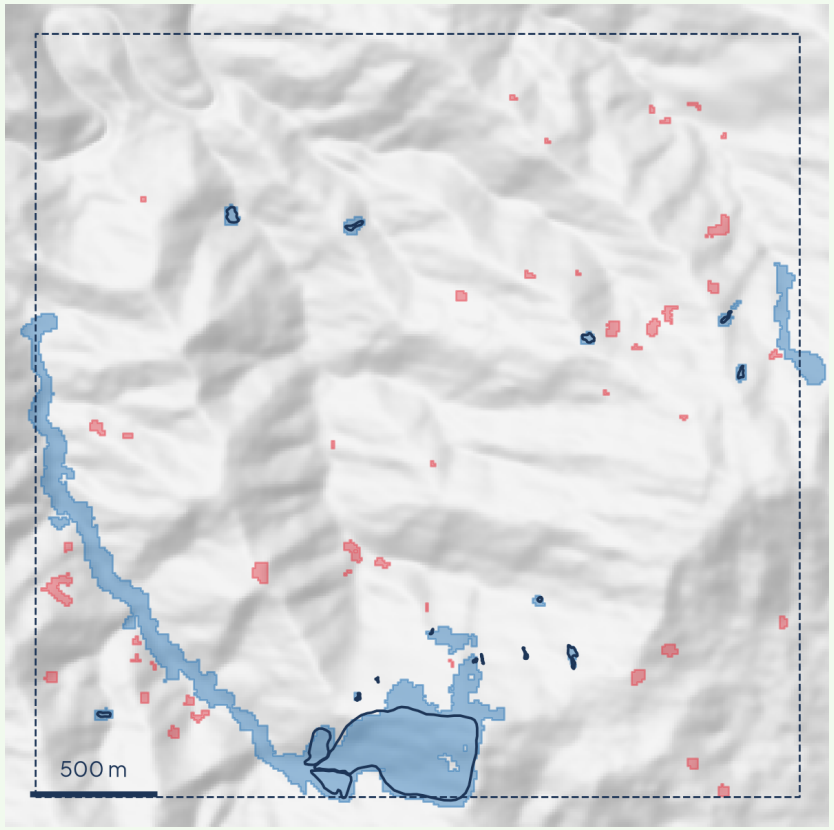
Polygon size	Chosen model	Simple logistic	Naive threshold
Under 5 px (outside the target)	35%	47%	45%
5 to 20 px	79%	85%	85%
Over 20 px	**98%**	**96%**	**96%**

Source: reports/tables/test_results.csv

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In the densest test block, flagged objects touch 10 of 10 scars of 500 m2 or more, and 44 other objects touch no scar

Block (6, 13) has the most inventory polygons (19) of the 34 test blocks. Its long blue object follows a river bed and counts as true because it touches a scar.



- inventory polygon
- inventory polygon, not touched
- flagged object touching a polygon
- flagged object touching none

Counts. 19 inventory polygons, 10 of them 500 m2 or more; 57 flagged objects, 13 touching a polygon.

Untouched polygons. 3 polygons are not touched by any object, all under 500 m2.

Reading it. False alarms here are small and scattered. The loose 'touches' rule also credits long objects, which is why strict recall is reported next to it.

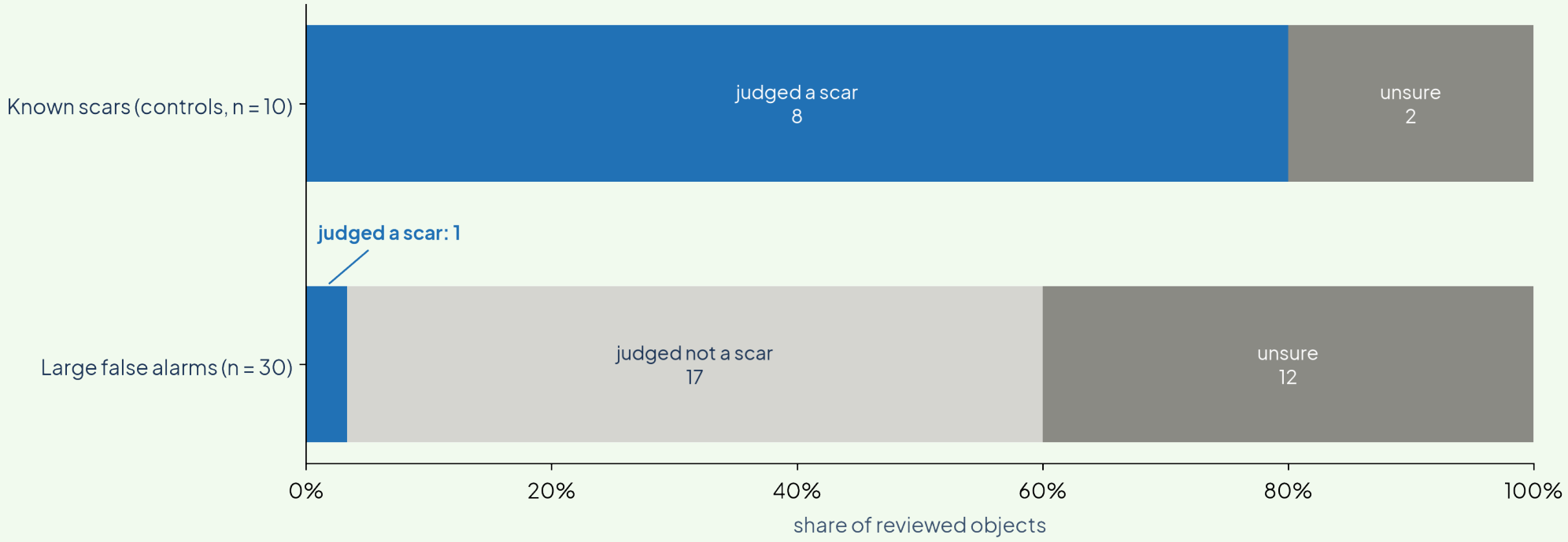
Note: inventory frame, UTM 49S; dashed line = block edge; objects counted by their representative point

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In a blinded review, 1 of 30 large false alarms was judged a scar, 17 were not and 12 were unsure

Reviewers recognised 8 of 10 known scars shuffled in as controls; the most common note on false alarms was terrace farming (7).

Blinded visual review on Sentinel-2 true colour, share of objects by verdict



Note: only objects of 20 px or more (24% of false alarms), n = 30; the reviewer saw the same imagery as the model

Source: blinded review answers (aggregated counts)

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Four follow-up ideas were tried after the blinded review; none gave a precision gain that held up, so none was kept

Precision at 80% recall in block cross-validation, first repeat; the size prior is the mean of 5 repeats.

Object-level second stage. 12.4% against 7.8% for the single stage, but only because objects merged into large blobs and shape recovery fell. A geometry guard was added to the validation.

Prior-year NDVI. 6.9% against 7.8% without it. Terrace false alarms have the same seasonal swing as scars.

Post-event persistence (2019, 2020). 6.6% against 7.8%. Scars here are back to pre-event NDVI within a year.

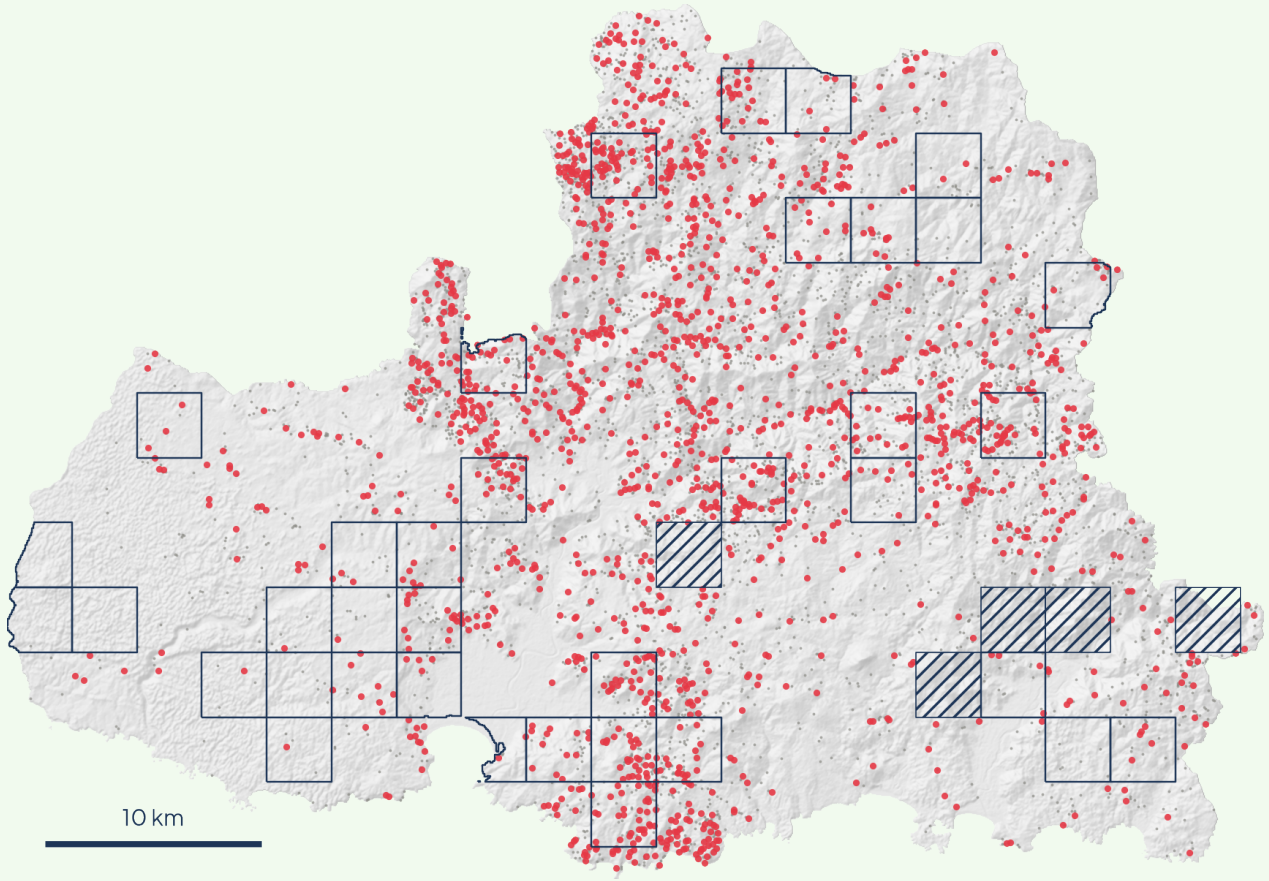
Object-size prior. Best rule 8.4% against 7.6% without it, a gain of 0.8 points whose interval includes zero. Not adopted.

Source: reports/tables/stage2_results.csv, prev_raw.csv, post_raw.csv, size_prior_results.csv

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The deliverable ranks 5,865 candidates across 1,370 km2; experts review the 1,678 large ones first

Each candidate carries a score and a blank verdict column, so expert answers can become new labels.



- large candidate (1,678)
- small candidate, 3 to 19 px (4,187)
- held-out test block
- excluded as likely unmapped

Order. Candidates are ranked by score: the 1,678 large ones (20 pixels or more) come first, then 4,187 small ones.

Fields. Score, size, slope, land use, distance to river and to the nearest mapped polygon, and empty verdict and note columns.

Layers. Two layers: on the Sentinel-2 grid, and shifted back to the inventory's frame.

Note: test objects: 1,154 in the candidate file (built over the whole area), 1,170 in the scored run (built over the test blocks only)

Labels: Samodra et al. 2024 (CC BY 4.0); Sentinel-2: Copernicus; embedding: Google DeepMind (CC BY 4.0)

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Read it as a screening aid: one event, 34 test blocks and an incomplete inventory limit what the numbers say

Observed: test scores. Assumed: label registration shift and land-use class names (inferred, unconfirmed by the inventory authors).

Inventory not wall-to-wall. It started from 328 citizen reports, so precision is a floor; the blinded review suggests this explains little of the gap for large objects.

Small test set. 34 blocks and 106 polygons of 500 m2 or more: intervals are wide and the point estimates would move with another split.

Calibration. The threshold was set for 80% recall on training blocks; test recall (86.8%) is above that, inside its interval.

Ten-metre imagery. Scars under about 500 m2 are not reliably visible, and the review used the same imagery as the model.

One event, one place. Nothing here says how the model transfers to another storm or region.

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Next: expert verdicts on the candidates become new labels; two data questions stay open

Nothing in this list is scheduled; it is what the evidence points to if the work continues.

Expert verdicts. Reviewing the large candidates and recording a verdict gives labels that do not depend on the original citizen reports.

Inventory footprint. The paper does not state the inventory's completeness and its author was not reachable; of five likely-unmapped blocks, one shows excess candidates.

Class names. Land-use and lithology class names are inferred from public maps and not confirmed by the authors.

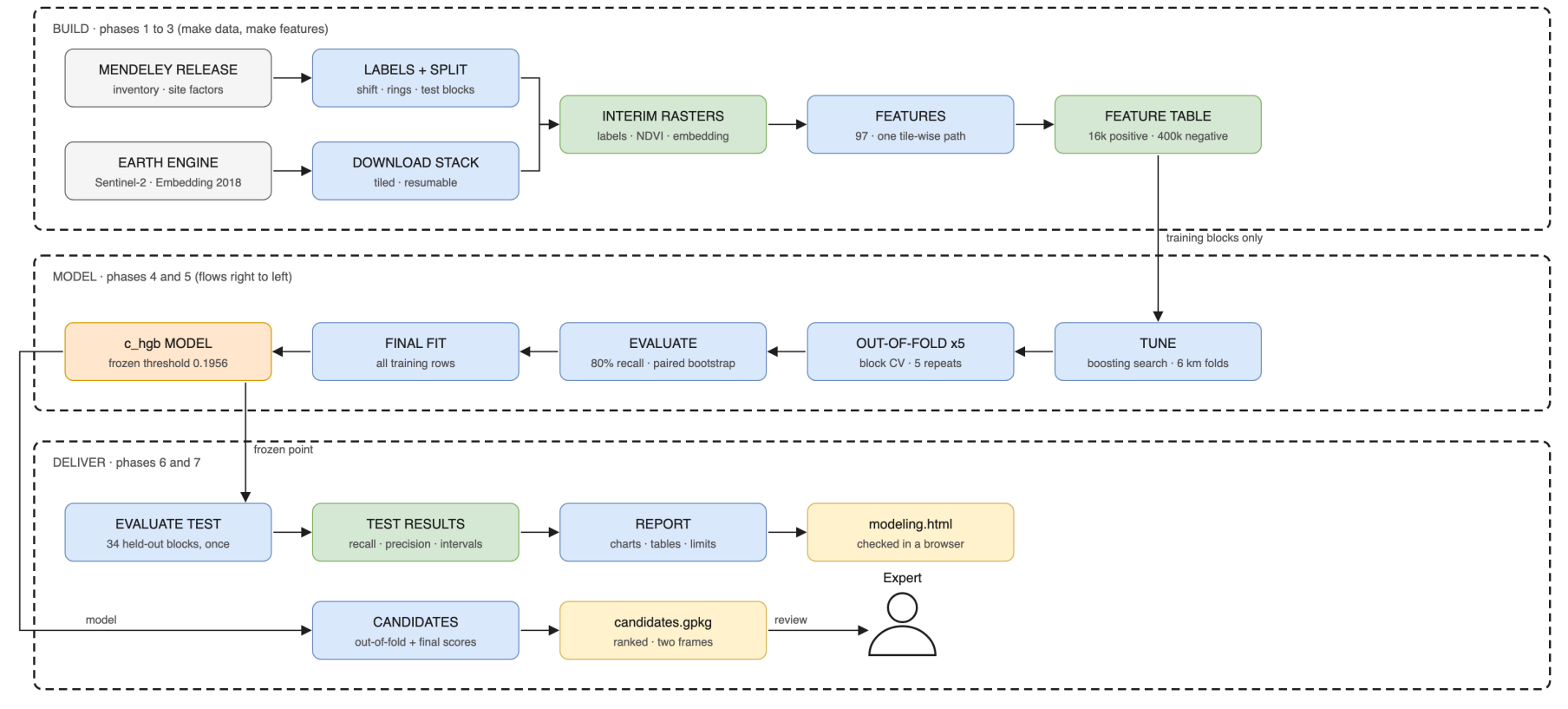
Dropped. A river-bank mask: scars sit near rivers and precision is highest there.

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Appendix: the pipeline runs from raw inventory and imagery to a frozen model, a once-only test score and a candidate file

Build, model and deliver phases; the test blocks are touched only in the deliver phase.

Architecture and data flow



Source: github.com/rlukafiardi/landslide-inventory-pacitan docs/architecture.svg

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Appendix: removing any one feature group moves precision at 80% recall by at most 1.9 points, and no interval excludes zero

Groups overlap in information and the intervals are wide: no group was shown to be needed, which is not the same as none mattering.

Variant	Precision at 80% recall	Change vs full	95% interval of change
Full model (all features)	**7.8%**	**–**	**–**
Without pixel NDVI features	6.7%	-1.1 pts	-2.1 to +1.3
Without neighbourhood NDVI features	5.9%	-1.9 pts	-4.2 to +0.7
Without terrain features	6.3%	-1.5 pts	-2.4 to +1.7
Without land use and lithology	8.2%	+0.4 pts	-1.5 to +2.5
Without distance to road	7.8%	+0.0 pts	-1.3 to +1.9
Without land use	7.1%	-0.7 pts	-2.7 to +1.3

Note: first cross-validation repeat; the 7.6% on page 10 is the mean of 5 repeats

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Appendix: the metrics in plain words

Recall and precision are explained on page 4; these are the other terms the results use.

Scar, polygon. A mapped landslide outline in the inventory. The targets count scars of 500 m2 or more (5 pixels or more at 10 m).

Flagged object. A connected group of at least 3 flagged pixels. A pixel is flagged when the model's score passes the threshold. Objects are what an expert reviews.

Object recall. Of the scars of 500 m2 or more, the share that at least one flagged object touches. In plain words: out of every 100 real scars, how many did the model find?

Object precision. Of the flagged objects, the share that touch a mapped scar. In plain words: out of every 100 flags, how many are real? It is a floor, because a flag on a real scar that nobody mapped counts as wrong.

Threshold, trade-off. A lower score threshold flags more objects: recall goes up and precision goes down. The threshold (the operating point) was fixed on training blocks before the test.

Strict recall (IoU). Recall that needs real overlap, not just a touch. IoU (intersection over union) is the overlap area divided by the combined area of object and scar; here at least 0.3.

False objects per true scar. Flagged objects that touch no scar, divided by the objects that touch one: how many flags an expert clears for each scar found.

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Appendix: method and data terms in plain words

Baseline. A simple method the model has to beat, such as a threshold on vegetation loss or a plain logistic regression.

Held-out test blocks. 3 km squares kept out of all training and tuning, and scored once at the end with the frozen model.

Block cross-validation. Train on some 3 km blocks, check on the others, rotate and repeat. Whole blocks are held out so that neighbouring pixels cannot leak the answer.

95% interval. The range the number could plausibly take given a small test set; here from resampling whole blocks (block bootstrap).

AUC. The chance that a random scar pixel scores higher than a random non-scar pixel: 0.5 is a coin flip, 1.0 is perfect. It hides how many false alarms a rare event produces, so models are compared at a fixed recall instead.

Ablation. Remove one group of features, refit and re-test, to see whether the group was needed.

Boosting, logistic regression. Boosting adds many small decision trees, each correcting the errors of the last. Logistic regression is a linear model that turns weighted features into a probability.

NDVI. A vegetation index from the satellite's red and near-infrared light: high is dense green, low is bare ground. dB is its change from before the storm to April-May 2018; a large drop means vegetation was lost.

Satellite Embedding. A 64-number summary per pixel, produced by a Google model from what the satellite saw during 2018. The individual numbers have no single physical meaning.

Hillshade. A shaded-relief image computed from the elevation model, used as a grey base for maps.

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About

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Related

https://github.com/rlukafiardi/landslide-inventory-pacitan

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